TOPOLOGY OF THE INVERSE LIMIT: THREE EQUIVALENT DESCRIPTIONS

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Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Opens of the limit. Limits, Lemma lemma-descend-opens (same chapter)

Source reference. Original lemma 01YX: sources/stacks-chow/tags.txt, source lines 2572–2572. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0A2Q: sources/stacks-chow/tags.txt, source lines 12713–12713. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0CUE: sources/stacks-chow/tags.txt, source lines 16099–16099. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01YW: sources/stacks-chow/tags.txt, source lines 2571–2571. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Let $S = \lim S_i$ be the limit of a directed inverse system of schemes with affine transition morphisms (Lemma 01YX). Then $S_{top} = \lim_i S_{i,top}$ where S_{top} indicates the underlying topological space of the scheme S.

Proof. We will use the criterion of Topology, Lemma 0A2Q. We have seen that $S_{set} = \lim_i S_{i,set}$ in Lemma 0CUE. The maps $f_i : S \to S_i$ are morphisms of schemes hence continuous. Thus $f_i^{-1}(U_i)$ is open for each open $U_i \subset S_i$. Finally, let $s \in S$ and let $s \in V \subset S$ be an open neighbourhood. Choose $0 \in I$ and choose an affine open neighbourhood $U_0 \subset S_0$ of the image of s. Then $f_0^{-1}(U_0) = \lim_{i \geq 0} f_{i0}^{-1}(U_0)$, see Lemma 01YX. Then $f_0^{-1}(U_0)$ and $f_{i0}^{-1}(U_0)$ are affine and

$$\mathcal{O}_S(f_0^{-1}(U_0)) = \operatorname{colim}_{i \geq 0} \mathcal{O}_{S_i}(f_{i0}^{-1}(U_0))$$

either by the proof of Lemma 01YX or by Lemma 01YW. Choose $a \in \mathcal{O}_S(f_0^{-1}(U_0))$ such that $s \in D(a) \subset V$. This is possible because the principal opens form a basis for the topology on the affine scheme $f_0^{-1}(U_0)$. Then we can pick an $i \geq 0$ and $a_i \in \mathcal{O}_{S_i}(f_{i0}^{-1}(U_0))$ mapping to a. It follows that $D(a_i) \subset f_{i0}^{-1}(U_0) \subset S_i$ is an open subset whose inverse image in S is $D(a)$. This finishes the proof. □

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